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Launcher of short projectiles with detachable barrel

Abstract

A toy projectile launcher having a housing, a projectile container, a cocking slide, an air piston assembly, a barrel interface section, and a detachable launch barrel is disclosed. The projectile container is adapted to hold projectiles, such as foam darts. The cocking slide can be moved forward and backward. When the cocking slide is moved backward the air piston barrel moves backward, and a projectile is moved in front of the air piston assembly. When the cocking slide is moved forward, the projectile is pushed into the barrel interface section by an air nozzle disposed on the front of the air piston assembly. A first airtight seal is formed between the air piston assembly and the rear of the foam dart, and a second airtight seal is formed between the barrel interface section and the detachable launch barrel.

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Background/Summary

REFERENCE TO OTHER APPLICATIONS (1) This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/151,853, filed on Feb. 22, 2021, entitled “LAUNCHER OF SHORT PROJECTILES WITH DETACHABLE BARREL,” the contents of which are incorporated by reference herein in their entirety.

FIELD

(1) The present invention is generally related to a toy projectile launcher, such as a toy pistol, gun, and the like, for launching toy projectiles, such as foam bullets, darts, balls, and the like, with a simplified construction and improved performance.

BACKGROUND

(2) Traditional toy projectile launchers have utilized various forms of rifles, pistols, blasters, machine guns, and the like, for launching toy projectiles, such as foam balls, darts, to name a few. Such toy launchers have varied in size, power, storage capacity, to name a few. More specifically, toy launchers of foam projectiles-bullets (or “darts”), balls, and the like have become ubiquitous. One standard for foam bullets has been marketed under the brand name Nerf® with a rubber tip and a foam body that totals approximately 71.5 mm in length. Other high-performance darts may be shorter in length. There have been various types of rifles, machine guns, and the like, that have been marketed for launching such foam projectiles.

(3) The caps of the toy darts are generally made of a material other than foam that allows the dart to

be shot from the launcher at a targeted person or object and/or propelled over an appropriate distance and/or at a relatively quick speed.

(4) Conventional dart guns have traditionally been marketed to pre-teen children for casual play. More recently, in conjunction with the advent of special event war games—such as paintball, laser tag, and the like—more high-powered launchers have been developed to target enthusiasts for such special events using foam darts.

(5) As an example, launchers having metal barrels, instead of plastic ones, have been used for improved launching velocity. Such launchers and darts are usually dimensioned to have a very small clearance—between the inner diameter of the barrel of the launcher and the outer diameter of the dart—so as to provide improved launching speed and accuracy.

(6) Accordingly, each launcher type has a specific design that accommodates either lower-powered, safer requirements for younger children or more high-powered requirements for older children or adults. This prevents such conventional launchers from being useful across all categories of play.

(7) With the above in mind, there is a need to improve the user adjustability of launchers in terms of safety, launching speed and accuracy.

SUMMARY

(8) To address the above, the present invention is generally related to an improved air-powered toy launcher for launching high performance foam darts. According to an exemplary embodiment of the present disclosure, a toy projectile launcher is equipped to provide a user the ability to transform the launcher between a configuration that complies with safety standards applicable to pre-teen users and a configuration that satisfies the high-power demands of older or professional users. To that end, a launcher may include a detachable launch barrel that forms an airtight seal at the front end of a foam dart so that the dart is launched with a higher launch force, and when the detachable launch barrel is removed or replaced with a slightly larger diameter detachable launch barrel that does not have a front seal, the front seal is eliminated, thereby resulting in a lower power launch force.

(9) In exemplary embodiments, the launcher may include a mechanism for priming a high-performance foam dart from a storage compartment into a firing position in a launch barrel while forming an airtight seal between an air piston nozzle and the launch barrel, thus improving the launch force on the primed dart. For example, co-pending U.S. patent application Ser. No. 16/906,996 and PCT Patent Application No. PCT/SG2021/050248 disclose respective launchers in pistol configurations that include a mechanism for priming darts stored in a storage handle or a cartridge into a firing position in the launch barrel of the pistol launchers. (The contents of U.S. patent application Ser. No. 16/906,996 and PCT Application No. PCT/SG2021/050248 are incorporated herein by reference in their entirety.) In the disclosed launchers, an air piston assembly is movable by a cocking slide in a two-step loading/priming motion, where the air piston assembly retracts—on a pull back on the cocking slide—to allow a top dart in the storage compartment (handle or cartridge) to be lifted to a position in front of the air piston assembly and where the air piston assembly pushes forward—on a push forward on the cocking slide—so that a front air nozzle of the air piston assembly pushes the top dart into the launch barrel and forms an airtight seal with the launch barrel behind the primed dart. It is anticipated that the detachable launch barrel of the present invention can be used with other configurations of air-powered launchers.

(10) In exemplary embodiments, the launcher may include a mechanism to accommodate an air piston nozzle that reaches through a storage drum to prime darts stored in the storage drum and to form an airtight seal with a launch barrel (or launch barrel interface section) in front of the storage drum. For example, co-pending PCT Application No. PCT/SG2021/050250 discloses a resilient mechanism that holds each foam dart in place in a storage area and that allows an air piston nozzle to reach through the storage area (behind the projectile) and to form a sealed connection with the launch barrel, thereby forming a rear seal behind the projectile. (The contents of PCT Application

No. PCT/SG2021/050250 are incorporated herein by reference.) Attachment of the detachable barrel creates a front seal for the projectile, and the combination of the back and front seals results in a higher-launch force. Without the detachable barrel, there is no front seal, and thus the projectile fires with less power. Similarly, a low power detachable launch barrel can be used which does not form a front seal and/or has a slightly larger internal diameter to reduce the launch force of the projectile. Such launch barrel may improve accuracy without increasing the velocity of the projectile.

(11) According to an exemplary embodiment, the toy launcher includes a housing, a projectile holder disposed within the housing and configured to contain a plurality of projectiles, a barrel interface section disposed at the front of the housing, an air piston assembly disposed within the housing. According to exemplary embodiments, the air piston assembly includes an air piston barrel having an air nozzle disposed on a front portion thereof, a plunger element, and a compression spring.

(12) According to an exemplary embodiment, the toy launcher includes a cocking slide that is adapted to be moved forward and backward relative to the housing, where the projectile holder and the air piston assembly are coupled to the cocking slide.

(13) According to an exemplary embodiment, the toy launcher includes a first detachable launch barrel that is adapted to (i) engage and attach to the barrel interface section; and (ii) disengage and detach from the barrel interface section.

(14) According to exemplary embodiments, when the cocking slide is moved backward from a forward position to a backward position, the air piston barrel moves backward and pushes the plunger element to compress the compression spring against the rear wall of the housing.

(15) According to exemplary embodiments, when the cocking slide is moved forward from the backward position to the forward position the air piston barrel moves forward, forming an internal air chamber between the front portion of the air piston barrel and the plunger element, and the air nozzle moves forward to form a first airtight seal between the air piston barrel and the barrel interface section.

(16) According to exemplary embodiments, when the first detachable launch barrel is attached to the barrel interface section, a second airtight seal is formed between the air nozzle and the first detachable launch barrel.

(17) According to exemplary embodiments, when the cocking slide is moved backward from a forward position to a backward position, a first projectile contained in the projectile holder is moved to a position in front of the air piston barrel.

(18) According to exemplary embodiments, when the cocking slide is moved forward from the backward position to the forward position, the air nozzle pushes the first projectile into the barrel interface section into a firing position in front of the first airtight seal and behind the second airtight seal.

(19) According to exemplary embodiments, when the first detachable launch barrel is detached from the barrel interface section, no airtight seal is formed in front of the first projectile.

(20) According to exemplary embodiments, the toy launcher further comprises a second detachable launch barrel, where the second detachable launch barrel is adapted to (i) engage and attach to the barrel interface section; and (ii) disengage and detach from the barrel interface section. According to exemplary embodiments, when the second detachable launch barrel is attached to the barrel interface section, no airtight seal is formed between the air nozzle and the second detachable launch barrel when the cocking slide is moved forward from the backward position to the forward position.

(21) According to exemplary embodiments, the toy launcher further comprises a latching assembly coupled between the plunger element and a trigger assembly, wherein the trigger assembly is adapted to be pulled backward by a user of the toy projectile launcher.

(22) According to exemplary embodiments, when the first projectile is in the firing position, the air

nozzle is immediately adjacent to the first projectile.

(23) According to exemplary embodiments, the toy launcher is configured as a pistol having a projectile holder disposed within a handle of the housing, In accordance with some of these embodiments, when the cocking slide is moved backward from the forward position to the backward position, the first projectile is lifted from the projectile holder disposed within the handle into the position in front of the air piston assembly.

(24) According to exemplary embodiments, the projectile holder is a rotatable storage drum. In accordance with some of these embodiments, when the cocking slide is moved backward from the forward position to the backward position, the projectile holder rotates to place the first projectile into the position in front of the air piston assembly.

(25) According to exemplary embodiments, the projectile holder includes a plurality of resilient projectile stoppers that each abut a portion of a respective projectile loaded in the projectile holder. In exemplary embodiments, each of the resilient projectile stoppers includes a surface disposed to face and be pushed by the air nozzle when the air piston barrel is pushed forward when the cocking slide is moved from the backward position to the forward position. In exemplary embodiments, each resilient projectile stopper, when pushed by the air nozzle, flexes outward to make way for the air nozzle to extend through the projectile holder.

(26) According to exemplary embodiments, the projectiles are foam darts.

(27) According to an exemplary embodiment, the toy launcher includes a coupling between the cocking slide and the air piston barrel.

(28) According to an exemplary embodiment, the projectile holder includes a projectile advancement mechanism for advancing a next loaded projectile in the projectile holder into a priming position in front of the air piston barrel.

(29) According to an exemplary embodiment, the second detachable launch barrel has a larger inside diameter than the first detachable launch barrel.

(30) According to an exemplary embodiment, the plunger element is pushed forward by the compression spring to expel the air from the internal air chamber through the front air nozzle on the front portion of the air piston barrel behind the next loaded projectile in the firing position when the coupling of the latching assembly between the plunger element and the trigger assembly is released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments of the present disclosure will be described with references to the accompanying figures, wherein:

(2) FIG. 1A is a schematic partial cross-sectional side views of key elements of a toy projectile launcher according to an exemplary embodiment of the present disclosure with a detachable launch barrel attached.

(3) FIG. 1B is a schematic partial cross-sectional side view showing attachment of a detachable launch barrel to the toy projectile launcher according to an exemplary embodiment of the present disclosure.

(4) FIG. 1C is a schematic partial cross-sectional side view showing removal of a detachable launch barrel from the toy projectile launcher according to an exemplary embodiment of the present disclosure.

(5) FIG. 1D is a schematic partial cross-sectional front view of a projectile storage drum useable with the toy projectile launcher of FIG. 1A according to an exemplary embodiment of the present disclosure.

(6) FIG. 2 is a schematic partial cross-sectional side view of the toy projectile launcher of FIG. 1A

according to an exemplary embodiment of the present disclosure with a detachable launch barrel removed.

(7) FIG. 3 is a schematic partial cross-sectional side view of the toy projectile launcher of FIG. 2 with a cocking slide being placed in a rearward loading and priming (cocked) position according to an exemplary embodiment of the present disclosure.

(8) FIG. 4 is a schematic partial cross-sectional side view of the toy projectile launcher of FIG. 2 with a cocking slide placed in a rearward loading and priming (cocked) position according to an exemplary embodiment of the present disclosure.

(9) FIG. 5 is a schematic partial cross-sectional side view of the toy projectile launcher of FIG. 4 the cocking slide being returned to a forward firing position according to an exemplary embodiment of the present disclosure.

(10) FIG. 6 is a schematic partial cross-sectional side view of the toy projectile launcher of FIG. 5 after a trigger pull illustrating the launch of a foam dart.

(11) FIG. 7A is a cross sectional side view of a detachable barrel according to an exemplary embodiment of the present invention.

(12) FIG. 7B is a cross sectional side view of a detachable barrel according to an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

(13) The present invention is generally related to an improved toy launcher that is modifiable by a user so that the launcher may operate with either higher power and more accuracy or lower power and more safety. To achieve this objective, according to an exemplary embodiment, a toy launcher incorporates a detachable launch barrel, so that when the launch barrel is attached it provides an airway seal with the front of the projectile, thereby improving power and accuracy. When the barrel is removed or replaced with another barrel that might be larger in diameter as compared to the high power barrel, the front airway seal is eliminated and the projectile launches at a lower power to address safety concerns. For example, FIG. 7A shows a detachable launch barrel **160** with an internal diameter **D1** and FIG. 7B shows a detachable launch barrel **161** with an internal diameter **D2** that is larger than the internal diameter **D1**. The second barrel may provide increased accuracy without requiring increased launch power.

(14) FIG. 1A is a schematic partial cross-sectional view of key elements of a toy projectile launcher **100** according to an exemplary embodiment of the present disclosure. For clarity and simplicity in illustrating the key elements and mechanisms of toy projectile launcher **100**, portions that are not necessary to understand the scope and the spirit of the present disclosure are not shown. One of ordinary skill in the art would readily understand the supporting elements needed to house and support the various illustrated elements, including those that facilitate the insertion and removal of drum **105** into and out of launcher **100**, with various design choices that would not depart from the spirit and scope of the present disclosure.

(15) FIG. 1A is a schematic side cross-sectional view of a projectile launcher **100** in an un-cocked position according to an exemplary embodiment of the present disclosure. As shown in FIG. 1A, projectile launcher **100** is shaped to resemble a pistol. In embodiments, launcher **100** may be in various other shapes and arrangements without departing from the spirit and the scope of the disclosure, such as, for example, a Thompson submachine gun. As illustrated in FIG. 1A, a reciprocating air piston assembly comprised of a barrel **101**, a plunger element **102**, and a front air nozzle **103** is located above a handle **104** and disposed within a housing **110** of the projectile launcher **100** behind a projectile holding drum **105**. According to an exemplary embodiment, barrel **101** of the air piston assembly has a generally rounded cylindrical or an oval cross-sectional shape and plunger element **102** is biased away from a back wall **107** of the rear part of launcher housing **110** by a compression spring **115**. The plunger element **102** incorporates a size and a shape that correspond with the cross-sectional shape of barrel **101** so as to form an airtight seal with an internal surface of barrel **101**. According to an exemplary embodiment of the present disclosure,

plunger element **102** may incorporate a resilient O-ring (made from a resilient material, such as a polymer)(not shown) to form an improved seal.

(16) As shown in FIG. **1A**, barrel **101** is coupled to a cocking slide **117** that is fittingly coupled to a track (not shown) incorporated in the housing **110** of launcher **100**. As will be described in further detail below, moving the cocking slide **117** back and forth primes the air piston assembly while feeding a foam dart for launch.

(17) As shown in FIG. **1A**, the barrel **101** is coupled to the drum **105** by a linear-to-rotational drive mechanism, such as a lead screw and nut arrangement **115**, so that linear motion of the barrel **101** (due to movement of the slide **117**) results in rotational movement of the drum **105**. As will be described in further detail below, drum **105** for holding projectiles-such as foam darts/bullets and the like-would be advanced by the linear motion of the barrel **101** such that a next projectile would be delivered to a firing position. Correspondingly, a spring-loaded stopper block (not shown) may be incorporated in the top portion of housing **110** for holding drum **105** into an aligned position when drum **105** is advanced via the barrel **101**. In alternative exemplary embodiments, the barrel **101** need not be coupled to the drum **105**, in which case the drum **105** may be rotated manually to deliver a projectile to a firing position.

(18) In embodiments, drum **105** may incorporate attachment elements (not shown) for detachably engaging corresponding elements (not shown) in launcher **100** for a rotatable joint that allows for rotating advancement of the drum **105** by slide **117**, with a stopper block (not shown) ensuring an aligned unitary advancement of drum **105** upon each pull on slide **117** by a user.

(19) In the illustrated embodiment, drum **105** is configured to shoot toy darts. Darts may be loaded into drum **105** before drum **105** is loaded into launcher **100** and/or darts may be loaded and/or refilled in drum **105** after drum **105** is loaded into launcher **100**. According to an exemplary embodiment, dart, generally designated by reference number **170**, has an elongate dart body **175** and a cap **180** that is affixed to the dart body. Dart body **175** has a substantially cylindrical shape and comprises a foam material, or the like, and cap **180** comprises a rubber material, or the like. In embodiments, dart **170** may have a total length, e.g., within a range of approximately 33 mm to 45 mm, such as 35 mm, 36 mm, 37 mm, or 40 mm, to name a few. Alternatively, the dart may have a length of 71.5 mm. although the present invention is described in the context of a foam bullet/dart launcher that utilizes shortened foam bullets/darts. Correspondingly, dart **170** has an outer cross-sectional diameter at its widest point of 12.9 mm. In alternative embodiments, dart **170** may have an outer cross-sectional diameter at its widest point of, for example, 12.5 mm, 13 mm, 14 mm, or 15 mm, to name a few. In embodiments, dart **170** may incorporate one or more recesses and corresponding ridges on its foam body—for example, as disclosed in U.S. patent application Ser. No. 16/895,172 filed on Jun. 8, 2020, the entire contents of which are incorporated by reference herein. Drum **105** is sized according to the darts it will utilize.

(20) As shown in FIG. **1A** and as will be described in further detail below, drum **105** incorporates an S-shaped cantilever stopper spring **140** for each corresponding projectile holder that is flexible downward to allow nozzle **103** of the air piston assembly to extend through drum **105** to form a seal with a firing tube **165** formed within a detachable launch barrel **160**. Removal of the launch barrel **160** (and the corresponding firing tube **165**) removes the seal between the nozzle **103** and the firing tube **165** so that the projectile is launched with lower power. In exemplary embodiments, as shown in FIGS. **1B** and **1C**, the launch barrel **160** may be removably attached to a barrel interface section **190** disposed at the front of the launcher **100** by a locking mechanism **195**. Although the locking mechanism **195** is shown in the figures as a twist-and-lock mechanism, it should be appreciated that the locking mechanism is not limited by this, and in other exemplary embodiments may be a screw mechanism, a plug-in attachment, or any other arrangement that allows for removable attachment of the launch barrel **160** to the barrel interface section **190**.

(21) According to an exemplary embodiment, spring **140** is composed of a resilient thermoplastic material. In embodiments, other suitable resilient materials may be used for providing the

flexibility to spring **140** needed to flex away from dart holders **205** (FIG. 1D) when nozzle **103** is extended through drum **105**, as shown in FIG. 1, and to return to an original configuration to serve as a back stop (or stopper) for darts **170** held in dart holders **205** (FIG. 4) in drum **105**.

Conventional drums utilize rigid retaining walls at the rear of the drum to retain darts within the drum. When the drums have openings to allow darts to be pushed forward for launching, such openings necessarily have a smaller cross-sectional area than the dart holders of such drums. In an embodiment of the present disclosure, since spring **140** is movable out of dart holder **205** when nozzle **103** moves through dart holder **205**, the diameter of air nozzle **103** can be maximized.

(22) As illustrated in FIGS. 1B and 1C, barrel interface section **190** is fixed to housing **110** and includes a rear opening for receiving nozzle **103** and an opposite front opening that connects to launch barrel **160**. As will be described in further detail below, when the launch barrel **160** is fitted to the launcher **100**, barrel interface section **190** forms an airtight connection between air piston nozzle **103** and launch barrel **160** and provides for receiving a primed projectile, such as foam dart **170**, into a primed position for launch. Accordingly, a user may pull back cocking slide **117** to move the air piston assembly—i.e., barrel **101**, plunger element **102**, and nozzle **103**—backward and to advance drum **105** in a first, pull-back, priming step and the user may then push cocking slide **117** forward, in a second priming step, to push barrel **101** and nozzle **103** forward. As will be described in detail below and shown in FIGS. 2-6, a dart **170** held in drum **105** is pushed forward into barrel interface section **190** (and partially into the firing tube **165** of the launch barrel **160**) by nozzle **103** and into a launch position in front of nozzle **103**, which forms an airtight seal with launch barrel **160** behind dart **170** since the outside diameter of the air nozzle **103** is essentially the same as the inside diameter of the dart holder **205** and the inside diameter of launch barrel **160**.

(23) FIG. 1D is a schematic partial cross-sectional front view of drum **105** shown in FIG. 1A according to an exemplary embodiment of the present disclosure. As shown in FIG. 1D, drum **105** includes integrated dart holders **205** around its outer circumference, each dimensioned to accommodate a foam dart **170** for use with launcher **100** and through which nozzle **103** is extended to connect to barrel interface section **190** in the arrangement shown in FIG. 1A. A drum incorporating a different number of dart holders **205** and/or a different number of rows of dart holders **205** as that show in FIG. 1D may be used without departing from the spirit and the scope of the present disclosure. An example of a drum incorporating plural rows of darts is disclose in co-pending U.S. patent application Ser. No. 17/038,106. (The contents of U.S. patent application Ser. No. 17/038,106 are incorporated herein by reference.)

(24) Operation of the toy launcher will now be described with reference to FIGS. 2-6. In each of FIGS. 2-6 toy launcher **100** is shown with launch barrel **160** removed.

(25) In FIG. 2, the toy launcher **100** is at rest with the nozzle **103** inserted through the drum **105**. FIGS. 3-4 show the toy launcher proceeding from the configuration shown in FIG. 2 to a first, pull back, priming step. Specifically, as previously mentioned, toy launcher **100** includes barrel **101** with a plunger element **102** that form an air piston assembly. Barrel **101** is coupled to the cocking slide **117** so that a user may pull back barrel **101** and plunger element **102** as part of the first, pull-back, priming step. As shown in FIGS. 3 and 4, spring **115** is compressed between plunger element **102** and back wall **107**. Advantageously, plunger element **102** starts at a position near a front portion of barrel **101**, as shown in FIG. 1, and, therefore, compression spring **115** may be fully compressed in the position illustrated in FIG. 4.

(26) According to an exemplary embodiment of the present disclosure, back wall **107** includes an aperture that allows a dome-shaped rod portion **305** (FIG. 4) to extend through and past another aperture **310** (FIG. 2) that is incorporated in a spring-loaded plate **315** that is, in turn, coupled to a trigger assembly **320**. When a user pulls cocking slide **117** backward, barrel **101**, plunger **102**, and rod portion **305** are pushed back as well. Plate **315** is coupled to a compression spring **325** that biases plate **315** downward towards the trigger assembly **320**. According to an exemplary embodiment of the disclosure, the leading edge of dome-shaped rod portion **305** is rounded and

when it is pushed backward, the rounded leading sloped edge pushes upward on a top edge of aperture **310** in plate **315**, compressing spring **325**, so that rod portion **305** can be pushed through aperture **310** from the front of plate **315** to clear an opposing back side of plate **315**. Once rod portion **305** is pushed sufficiently past plate **315** through aperture **310**, spring **325** moves plate **315** downward into engagement with a notch or recess **330** (FIG. 4) opposite the rounded face of rod portion **305** so that rod portion **305**—and, correspondingly, plunger element **102**—is engaged with, and temporarily retained in place by plate **315**. As shown in FIG. 4, the notch **330** hooks to the opposing back side of plate **315** above aperture **310** once plate **315** is pushed downwardly by compression spring **325** into notch **330** and, accordingly, a top edge of aperture **310** is pushed into a bottom surface of notch **330**. Thus, plate **315**, compression spring **325**, and notch **330** together form a latching assembly for holding rod portion **305** in the backward position with compression spring **115** fully compressed.

(27) In alternative embodiments, a structural stop (not shown) may be used to limit the backward motion of cocking slide **117** to the above full extension position—i.e., the engagement position between notch **330** and plate **315**.

(28) Correspondingly, with barrel **101** and cocking slide **117** moved back to the configuration shown in FIG. 4, nozzle **103** is pulled back away from barrel interface section **190** and from one of the dart holders **205** in drum **105** through a rear opening of dart holder **205**, thus clearing the way on the rear end for drum **105** to rotate.

(29) In substantial synchronization with nozzle **103** being retracted from dart holder **205** of drum **105**, drum **105** is rotated to advance to a next dart holder **205**. As discussed above, according to an exemplary embodiment, the linear motion of the barrel **101** is translated into rotational motion imparted to the drum **105** through operation of a linear-to-rotational drive mechanism.

(30) FIG. 4 shows an individual dart holder **205** holding a dart **170** after drum **105** has been rotated upon a pull back on cocking slide **117**. In exemplary embodiments, each dart holder includes a main central portion, which is formed in the shape of a cylinder with a cross-sectional diameter of about 13 mm for fitting and holding the widest point(s) of the foam body of dart **170**. Each holder **205** includes an inner rear end opening for spring **140** to extend inward so that a hook element thereof forms a front facing flat surface stop to stop the rear end of dart **170** and thereby holding dart **170** in the main central portion of dart holder **205** and preventing movement of the dart **170** through the rear of the drum during loading. Each hook element serves to abut the rear end of each corresponding dart that is loaded into drum **105** by insertion through a front end of each dart holder **205**. According to an exemplary embodiment, launch barrel **160** and barrel interface section **190** each have an inner diameter of approximately 13.26 mm to provide minimal clearance for dart **170**, which each has an outer diameter of approximately 13 mm. According to an exemplary embodiment, main portion, including hook portion, has an interior diameter of about 12.9 mm and may be tapered slightly from hook element to front end—in other words, having a slightly larger interior circumference towards front end—to allow for inserting each dart **170** from front end to abut front facing surface of hook element and for holding each dart **170** in place. As an example, the interior diameter of main portion near front end is slightly more than 12.9 mm and the interior diameter of main portion near hook element is slightly less than 12.9 mm. The dimensions provided herein are merely exemplary and not intended to limit the invention in anyway.

(31) Referring now to FIG. 5, with the notch/recess **330** of rod portion **305** engaged with plate **315** via the downward bias of spring **325**, the user can push cocking slide **117** forward in a second priming step—see forward arrow adjacent cocking slide **117** in FIG. 5. Consequently, barrel **101** is pulled forward towards the front of launcher **100** while rod portion **305** and plunger element **102** are held in place by plate **315**. As shown in FIG. 5, compression spring **115** remains fully compressed by the return of cocking slide **117** to its original forward position. Accordingly, plunger element **102** forms an air chamber **405** within barrel **101** whereby air is drawn in through nozzle **103** of barrel **101**. In accordance with an exemplary embodiment of the present disclosure, barrel

101 incorporates a resilient element (not shown) on a rear internal surface thereof to further improve the seal for air chamber **405** and to provide cushioning between the front surface of plunger element **102** and the rear internal surface of barrel **101**. Nozzle **103** may be of a substantially smaller diameter than that of the air chamber **405** so that a forward push by plunger **102** would expel the air through nozzle **103** at a higher pressure.

(32) As further shown in FIG. 5, as the cocking slide **117** is moved forward in the direction shown by the forward arrow, a dart **170** is pushed forward from drum **105** into barrel interface section **190** (and, as shown in FIG. 1A, when the launch barrel **160** is installed on the launcher **100**, partially into the firing tube **165**) by nozzle **103**, now inserted back into the rear opening of barrel interface section **190** through dart holder **205**. The front portion of the nozzle **103** pressed into contact with the back portion of the dart **170** results in a rear airtight seal. When the launch barrel **160** is installed on the launcher **100** before the priming steps, a front airtight seal is also formed between the dart **170** and the firing tube **165**. The combination of the front and back seals provides the dart **170** with a higher velocity flight as compared to that achieved without the launch barrel **160** installed.

(33) As further shown in FIG. 5, upon insertion of the nozzle **103** into the dart holder **205**, the corresponding spring element **140** is flexed downward to make way for the nozzle **103**. As spring element **140** is moved completely out of the dart holder **205**, nozzle **103** can have nearly the same outside diameter as the inside diameter of the dart holder **205** thereby maximizing the diameter of the nozzle **103**, in contrast to smaller nozzles for fitting through conventional drums having a reduced diameter due to the use of rigid retaining walls at the rear of a drum.

(34) In an exemplary embodiment, nozzle **103** incorporates an O-ring **303** around its outer circumference to form a seal around the internal circumference of the rear opening of barrel interface section **190**. Advantageously, the rear airtight seal is formed from air chamber **405** directly through nozzle **103** to the rear end of dart **170**, now placed in the firing position in barrel interface section **190**, without the need for any connections involving dart holder **205** and, thus, further improving the airtight connection. Additionally, spring **140** allows for nozzle **103** to be larger in cross section—for example, having an outer diameter of slightly less than 12.9 mm to fit through main portion **220** of each dart holder **205**—so that the rear airtight seal can be formed with barrel interface section **190** while having an air exit for nozzle **103** that substantially overlaps the rear end of dart **170**, thus improving the launch force on dart **170**.

(35) FIG. 6 shows a trigger pull resulting in the launch of dart **170** from the launcher **100**. In this regard, in accordance with an exemplary embodiment, the trigger assembly **320** includes a camming surface **325**. When the trigger assembly **320** is pulled backward by the user, locking plate **315** is caused to move upward by the camming surface **325**. In embodiments, trigger assembly **320** may be biased forward in a default position by a spring **350**, or the like, such that plate **315** returns to its downward position when the trigger assembly **315** is in the forward, default, non-firing position. As plate **315** is pushed upward by the camming surface **325** of trigger assembly **320**, the engagement between plate **315** and notch/recess **330** of rod portion **305** is released as aperture **310** is moved upward to a position that clears notch/recess **330**. Thus, as illustrated in FIG. 6, spring **115** is released from its fully compressed state thereby driving plunger element **102** forcefully forward until the front surface of plunger element **102** abuts the rear internal surface of barrel **101** to thereby expel the collected air from air chamber **405** through nozzle **103** to launch dart **170**. If the launch barrel **160** is not installed, the dart **170** is fired directly from the barrel interface section **190**, with just a rear airtight seal (between the nozzle **103** and the dart **170**) providing the launch force. With the launch barrel **160** installed, the dart **170** is fired through the launch barrel **170**, with the front airtight seal (between the dart **170** and the firing tube **165**) providing a front airtight seal in addition to the back airtight seal already provided between the nozzle **103** and dart **170**, thereby launching the dart **170** with higher velocity and accuracy as compared to that provided by the launcher **100** without the launch barrel **160**.

(36) After release of the trigger, trigger assembly **320** is returned to the forward default position and plate **315** is returned to its lowered position. According to an exemplary embodiment of the present disclosure, cocking slide **117** may be pulled backward again to the position shown in FIG. **4** to prime a next dart **170** in drum **105** and into the firing position shown in FIG. **5** by a push forward again on cocking slide **117**.

(37) Although the exemplary embodiment is described in the context of a foam bullet/dart launcher that utilizes shortened foam bullets/darts, it is to be understood that the two-step priming/loading and firing action according to the present disclosure could be applied to a toy projectile launcher of other types of projectiles (e.g., a ball or the like) or a fluid launcher whereby the fluid from a reservoir in the handle is driven by a plunger. In such environment the two-step priming/pumping action of the present disclosure enables a handheld high-velocity fluid burst launcher.

(38) While particular embodiments of the present disclosure have been shown and described in detail, it would be obvious to those skilled in the art that various modifications and improvements thereon may be made without departing from the spirit and scope of the disclosure. It is therefore intended to cover all such modifications and improvements that are within the scope of this disclosure.

Claims

1. A toy projectile launcher, comprising: a housing; a projectile holder disposed within the housing and configured to contain a plurality of projectiles; a barrel interface section disposed at a front portion of the housing; an air piston assembly disposed within the housing and including an air piston barrel having an air nozzle disposed on a front portion thereof, a plunger element, and a compression spring; a cocking slide that is adapted to be moved forward and backward relative to the housing, the projectile holder and the air piston assembly being coupled to the cocking slide; and a first detachable launch barrel that is adapted to (i) engage and attach to the barrel interface section; and (ii) disengage and detach from the barrel interface section; wherein, when the cocking slide is moved backward from a forward position to a backward position, the air piston barrel moves backward and pushes the plunger element to compress the compression spring against a rear wall of the housing; wherein, when the cocking slide is moved forward from the backward position to the forward position: the air piston barrel moves forward, forming an internal air chamber between a front portion of the air piston barrel and the plunger element; and the air nozzle moves forward to push one of the plurality of projectiles from the projectile holder into the barrel interface section to form a first airtight seal between the air piston barrel and the barrel interface section; and wherein, when the first detachable launch barrel is attached to the barrel interface section, a second airtight seal is formed between the air nozzle and the first detachable launch barrel.
2. The toy projectile launcher of claim 1, wherein, when the cocking slide is moved backward from a forward position to a backward position, a first projectile contained in the projectile holder is moved to a position in front of the air piston barrel.
3. The toy projectile launcher of claim 2, wherein the projectile holder includes a projectile advancement mechanism for advancing the first projectile contained in the projectile holder into the position in front of the air piston barrel.
4. The toy projectile launcher of claim 2, wherein, when the cocking slide is moved forward from the backward position to the forward position, the air nozzle pushes the first projectile into the barrel interface section into a firing position in front of the first airtight seal and behind the second airtight seal.
5. The toy projectile launcher of claim 4, wherein, when the first detachable launch barrel is detached from the barrel interface section, no airtight seal is formed in front of the first projectile.
6. The toy projectile launcher of claim 4, further comprising a latching assembly coupled between the plunger element and a trigger assembly, wherein the trigger assembly is adapted to be pulled

backward by a user of the toy projectile launcher.

7. The toy projectile launcher of claim 6, wherein the plunger element is pushed forward by the compression spring to expel air from the internal air chamber through the air nozzle behind the first projectile in the firing position when the coupling of the latching assembly between the plunger element and the trigger assembly is released.

8. The toy projectile launcher of claim 4, wherein, when the first projectile is in the firing position, the air nozzle is immediately adjacent to the first projectile.

9. The toy projectile launcher of claim 4, wherein the housing is configured as a pistol, and wherein the projectile holder is disposed within a handle of the housing.

10. The toy projectile launcher of claim 9, wherein, when the cocking slide is moved backward from the forward position to the backward position, the first projectile is lifted from the projectile holder disposed within the handle into the position in front of the air piston assembly.

11. The toy projectile launcher of claim 4, wherein the projectile holder is a rotatable storage drum, and wherein, when the cocking slide is moved backward from the forward position to the backward position, the projectile holder rotates to place the first projectile into the position in front of the air piston assembly.

12. The toy projectile launcher of claim 11, wherein the projectile holder includes a plurality of resilient projectile stoppers that each abut a portion of a respective projectile loaded in the projectile holder.

13. The toy projectile launcher of claim 12, wherein each of the resilient projectile stoppers includes a surface disposed to face and be pushed by the air nozzle when the air piston barrel is pushed forward when the cocking slide is moved from the backward position to the forward position.

14. The toy projectile launcher of claim 13, wherein each resilient projectile stopper, when pushed by the air nozzle, flexes outward to make way for the air nozzle to extend through the projectile holder.

15. The toy projectile launcher of claim 1, further comprising a second detachable launch barrel, wherein the second detachable launch barrel is adapted to (i) engage and attach to the barrel interface section; and (ii) disengage and detach from the barrel interface section; and wherein, when the second detachable launch barrel is attached to the barrel interface section, no airtight seal is formed between the air nozzle and the second detachable launch barrel when the cocking slide is moved forward from the backward position to the forward position.

16. The toy projectile launcher of claim 15, wherein the first detachable launch barrel has a first inside diameter and the second detachable launch barrel has a second inside diameter, and wherein the first inside diameter is less than the second inside diameter.

17. The toy projectile launcher of claim 1, wherein the projectiles are foam darts.

18. A toy projectile launcher, comprising: a housing; a projectile holder disposed within the housing and configured to contain a plurality of projectiles; a barrel interface section disposed at a front portion of the housing; an air piston assembly disposed within the housing and including an air piston barrel having an air nozzle disposed on a front portion thereof, a plunger element, and a compression spring; a cocking slide that is adapted to be moved forward and backward relative to the housing, the projectile holder and the air piston assembly being coupled to the cocking slide; a latching assembly coupled between the plunger element and a trigger assembly, wherein the trigger assembly is adapted to be pulled backward by a user of the toy projectile launcher; and a first detachable launch barrel that is adapted to (i) engage and attach to the barrel interface section; and (ii) disengage and detach from the barrel interface section; wherein, when the cocking slide is moved backward from a forward position to a backward position: the air piston barrel moves backward and pushes the plunger element to compress the compression spring against a rear wall of the housing; and a first projectile contained in the projectile holder is moved to a position in front of the air piston barrel; wherein, when the cocking slide is moved forward from the backward

position to the forward position: the air piston barrel moves forward, forming an internal air chamber between a front portion of the air piston barrel and the plunger element; and the air nozzle moves forward to push one of the plurality of projectiles from the projectile holder into the barrel interface section to form a first airtight seal between the air piston barrel and the barrel interface section with the first projectile disposed between the air piston barrel and the barrel interface section; wherein, when the first detachable launch barrel is attached to the barrel interface section: a second airtight seal is formed between a front portion of the first projectile and the first detachable launch barrel; and the air nozzle pushes the first projectile into the barrel interface section into firing position in front of the first airtight seal and behind the second airtight seal; and wherein the plunger element is pushed forward by the compression spring to expel air from the internal air chamber through the air nozzle behind the first projectile in the firing position when the coupling of the latching assembly between the plunger element and the trigger assembly is released.
